1.4. Subcellular organelles

An eukaryotic cell does not have a homogeneous internal environment but is divided into two major compartments ,cytoplasm and nucleus and subsequently into individual compartments, each of which is surrounded by a membrane, addressed as organelles.

An organelle can be specific for the type of cell

Certain organelles like lysosomes are present in animal cells but not in plant cells. Plant cells have chloroplasts, the sites of photosynthesis and usually a large, water filled vacuole. While most of the animal cells are surrounded by a plasma membrane only, plant cells often have a rigid cell wall on the outer membrane. Plants do not have centrioles. certain cells have basal bodies, which act as anchors.

1.4.1.Cell Membrane

All plants, animal cells, prokaryotic cells, and fungal cells are bounded by a cell membrane, which is sometimes known as plasma membrane.

Chemical composition of Cell membrane: Cellular membranes including plasma membranes and internal membranes consist of mainly lipid, protein and water. Lipids constitute about 40 percent of the membrane composition. Lipids are complex mixtures of cholesterol and fatty acid esters, mainly in the form of glycerides and phospholipids. Glycerol is a three-carbon molecule that forms the backbone of membrane lipids. Within an individual glycerophospholipid, fatty acids are attached to the first and second carbons, and the phosphate group is attached to the third carbon of the glycerol backbone.

Lipid bilayer encircles a cell and is amphipathic with one end as a hydrophilic 'head' and the other end as a hydrophobic 'tail'. Each 'leaf' of the lipid bilayer has one side consisting of an array of the hydrophilic heads, while the other side consists of the hydrophobic tails. An aqueous environment causes the hydrophobic tails to aggregate, so that the hydrophobic sides of each leaf come together to form a non-ionic centre, like an oil drop in water. The hydrophilic end of the two leaves face into the ionic milieu on either side of the lipid bilayer. The lipid bilayer has the important property of fluidity which allows it to fuse with other membranes, generate new membranes by fission, and provide solvent for proteins that can reside within the layer and move around within it. It can permit water but will not permit ions, small charged molecules, and all large molecules.

Unit 1.indd 8 21-05-2018 15:07:36